GLOWTAP

# Database Performance

Neon Postgres · 4/8/2026, 3:13:15 am IST · 5 measured rounds · DB 463 MB

26 workloads | 7 excellent | Latency suite | Health score 55%

| TESTS | EXCELLENT | GOOD | ACCEPTABLE |
|---|---|---|---|
| 26 | 7 | 5 | 14 |
| timed queries | | | |

| SLOW | DB SIZE | LIVE ROWS | NEON CU |
|---|---|---|---|
| 0 | 463 MB | 12,11,629 | ~0.25 |
| | | approx · all schemas | ~1 GB RAM · 1 CPU |

## Neon compute

Database host resources (not the laptop runner)

| METRIC | VALUE |
|---|---|
| Provider | Neon Postgres (serverless) |
| Region | ap-southeast-1 |
| Project ID | little-feather-12343290 |
| Endpoint ID | ep-lively-shadow-ao184du5 |
| Host | ep-lively-shadow-ao184du5-pooler.c-2.ap-southeast-1.aws.neon.tech |
| Estimated CU | ~0.25 CU |
| Estimated RAM | ~1 GB |
| Local file cache | ~0.75 GB |
| Allocated CPUs (num_cpus) | 1 |

| METRIC | VALUE |
| --- | --- |
| Estimate basis | shared_buffers ≈ 230 MB |

Hardware metrics are Neon compute (not the laptop that ran Node). num_cpus()=1 (fractional CUs may round up). Wall-clock timings still include client↔Neon WAN RTT.

## PostgreSQL settings

Neon serverless Postgres knobs

| METRIC | VALUE |
| --- | --- |
| Database | superAdmin |
| Version | 18.4 (df16b3c) |
| shared_buffers | 230 MB |
| work_mem | 4096 kB |
| effective_cache_size | 6553 MB |
| max_parallel_workers | 8 |
| max_parallel_workers_per_gather | 2 |
| max_connections | 901 |

## Dataset under test

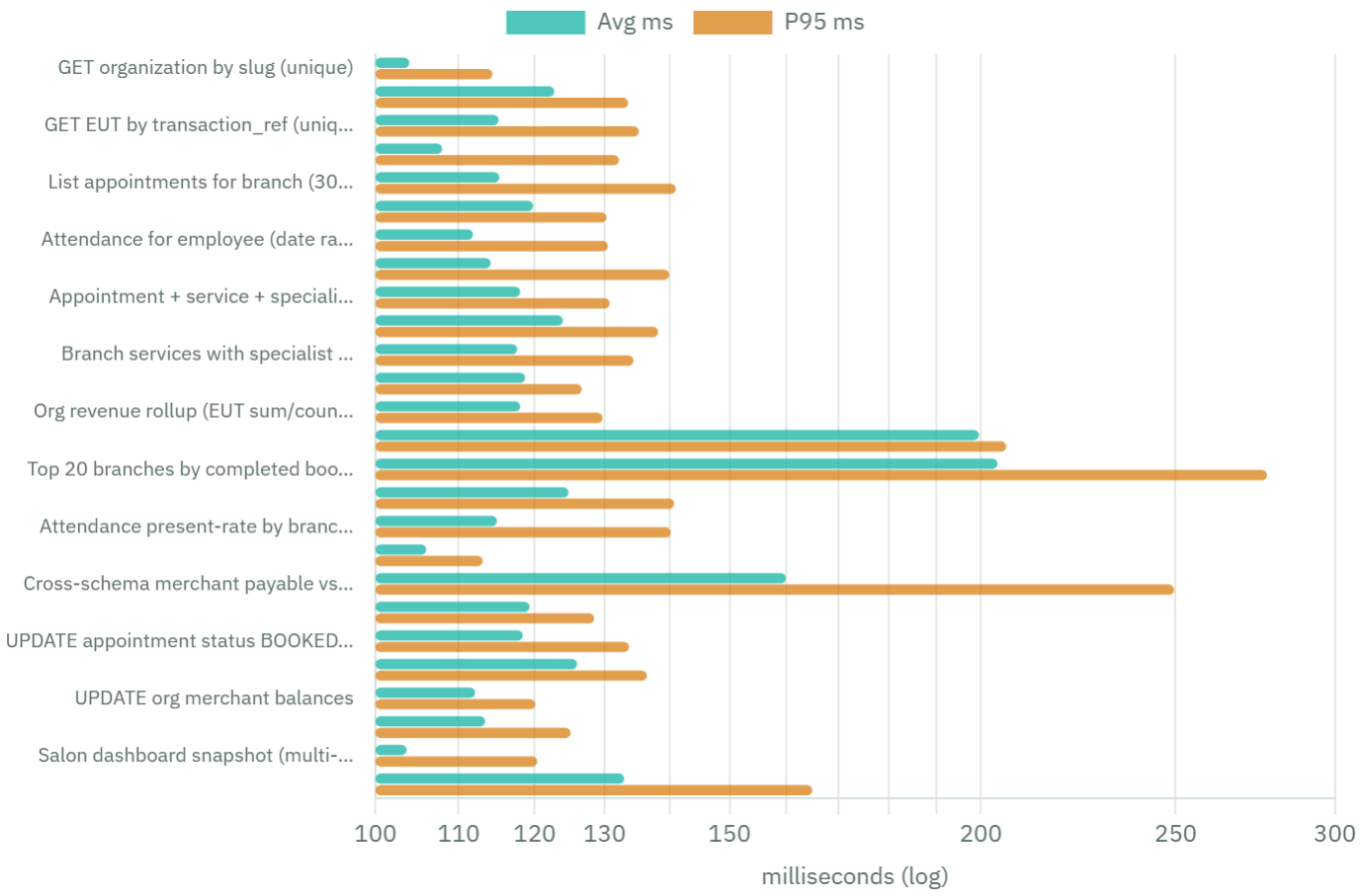
PERF seed + tracked live rows

| METRIC | VALUE |
| --- | --- |
| Database size | 463 MB |
| Approx. live rows | 12,11,629 |
| PERF seed row sum | 7,57,350 |
| Organizations | 850 |
| Branches | 4,250 |
| Appointments | 1,70,000 |
| Status events | 1,70,000 |

| METRIC | VALUE |
|---|---|
| End-user transactions | 63,750 |
| Attendance rows | 2,72,000 |
| Employees | 34,000 |
| Branch services | 42,500 |

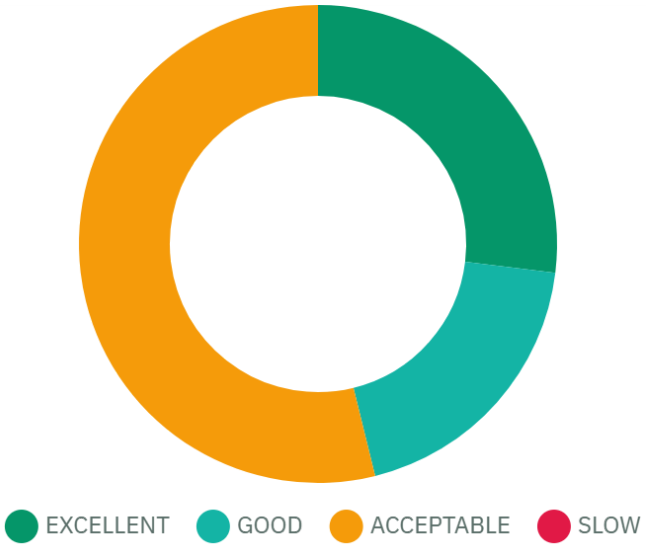
## Latency spectrum

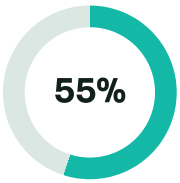
Avg vs p95 across every benchmark (log scale)




## Grade mix

Share of results under local SLO bands



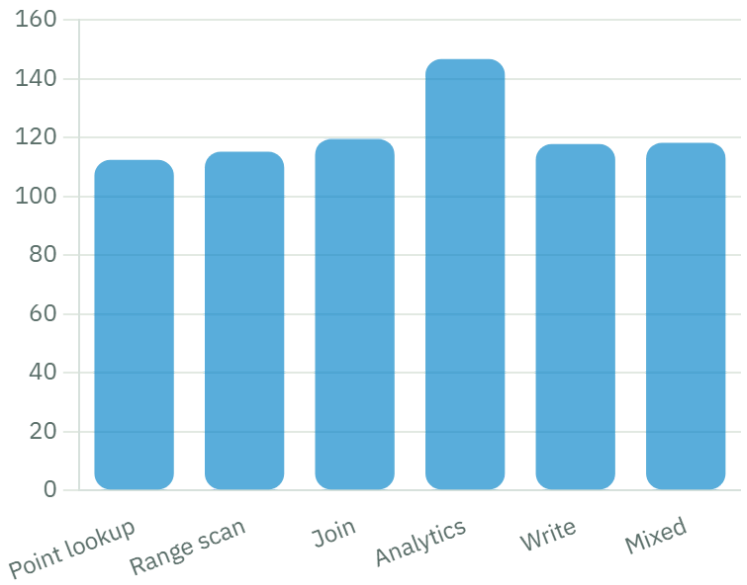




## Suite health

Weighted from grades (Excellent=4 … Slow=0). Target: keep analytics under 1s at this scale.

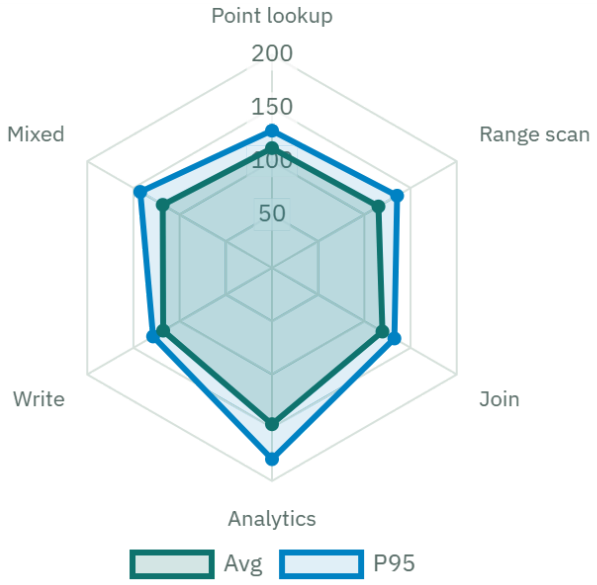
## By category

Average latency



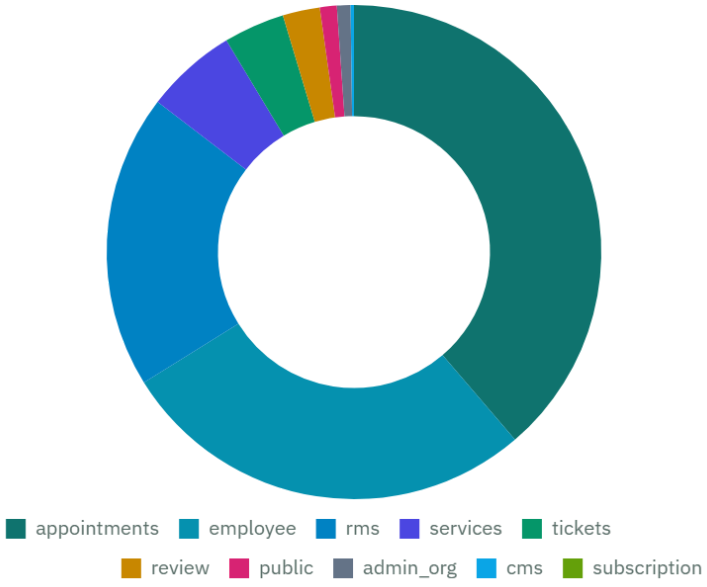

## Category radar

Avg vs p95 profile



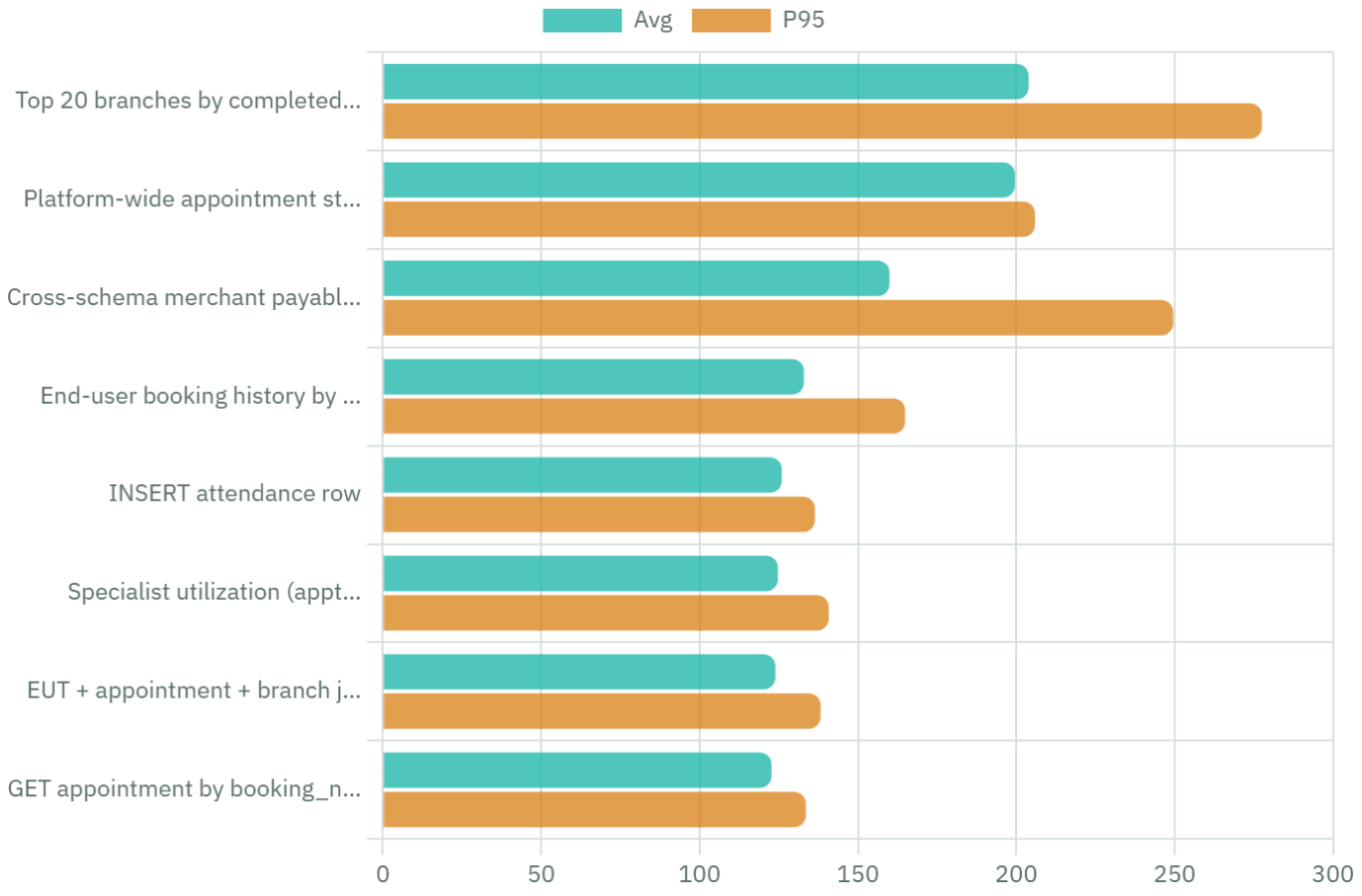

## Storage by schema

Disk footprint (MB)



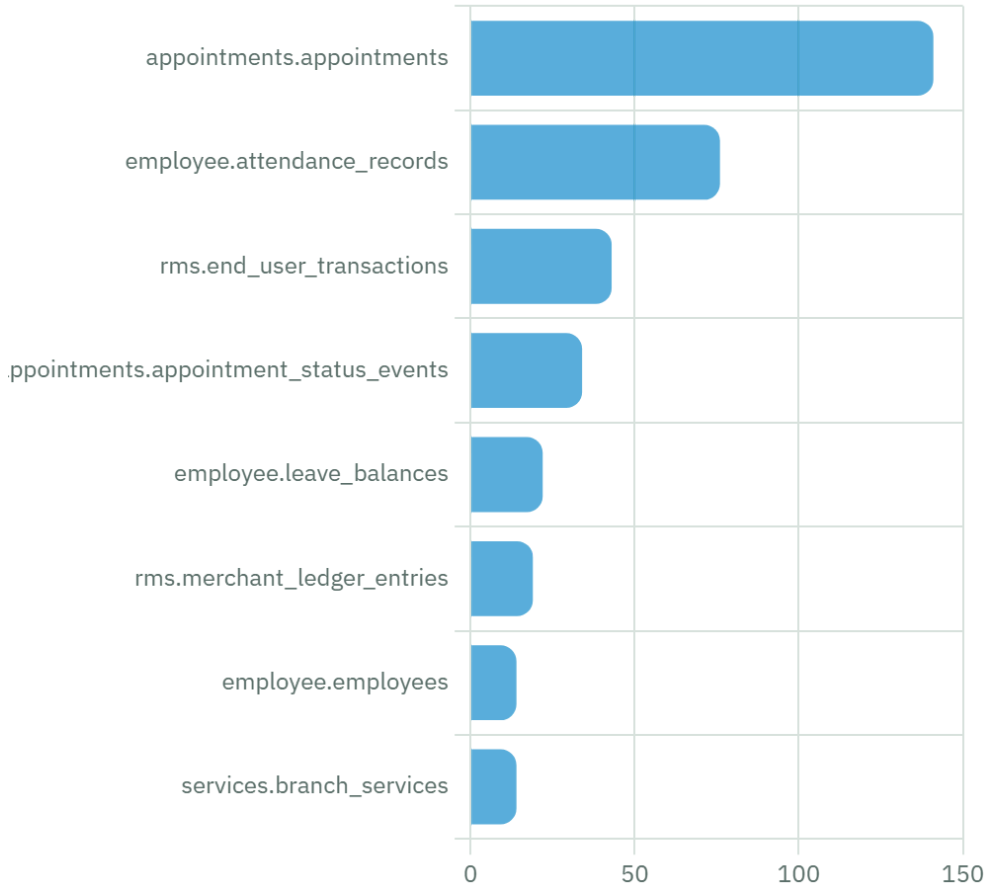

## Slowest queries

Highest avg ms after warm-up




## Largest tables

Live rows × size



## Results matrix

All timings in milliseconds · sticky header for scanning

| TEST | CATEGORY | ROWS | AVG | P50 | P95 | MAX | | |
|---|---|---|---|---|---|---|---|---|
| GET organization by slug (unique) | Point lookup | 1 | 103.95 | 100.4 | 114.34 | 114.34 | — | ACCEPT |
| GET appointment by booking_number (unique) | Point lookup | 1 | 122.72 | 120.66 | 133.57 | 133.57 | — | ACCEPT |
| GET EUT by transaction_ref (unique) | Point lookup | 1 | 115.14 | 119.29 | 135.2 | 135.2 | — | ACCEPT |
| GET employee by org + employee_number | Point lookup | 0 | 107.93 | 102.01 | 132.17 | 132.17 | — | ACCEPT |
| List appointments for branch (30-day window) | Range scan | 0 | 115.24 | 115.62 | 141.05 | 141.05 | — | ACCEPT |
| Filter appointments by status + org | Range scan | 0 | 119.78 | 120.77 | 130.28 | 130.28 | — | ACCEPT |
| Attendance for employee (date range) | Range scan | 0 | 111.81 | 109.99 | 130.5 | 130.5 | — | ACCEPT |

## Hottest relations

Top tables by on-disk size

| RELATION | LIVE ROWS | SIZE |
|---|---|---|
| appointments.appointments | 1,70,003 | 141 MB |
| employee.attendance_records | 2,72,002 | 76 MB |
| rms.end_user_transactions | 63,761 | 43 MB |
| appointments.appointment_status_events | 1,70,006 | 34 MB |
| employee.leave_balances | 1,02,006 | 22 MB |
| rms.merchant_ledger_entries | 63,763 | 19 MB |
| employee.employees | 34,002 | 14 MB |

| RELATION | LIVE ROWS | SIZE |
|---|---|---|
| services.branch_services | 42,512 | 14 MB |
| services.branch_service_specialists | 85,001 | 13 MB |
| tickets.support_tickets | 17,002 | 10208 kB |

## Methodology

Client wall time via `perf_hooks` around `pg` queries. One warm-up discarded; stats from 5 rounds (min / avg / p50 / p95 / p99 / max). Covering + partial indexes (`076` / `077`), range predicates, and LATERAL aggregates for platform analytics. Local SLO bands are tighter than Neon (no WAN RTT).

▶ **Full markdown appendix**

GlowTap — Database Performance · generated by scripts/run-perf-benchmark.js